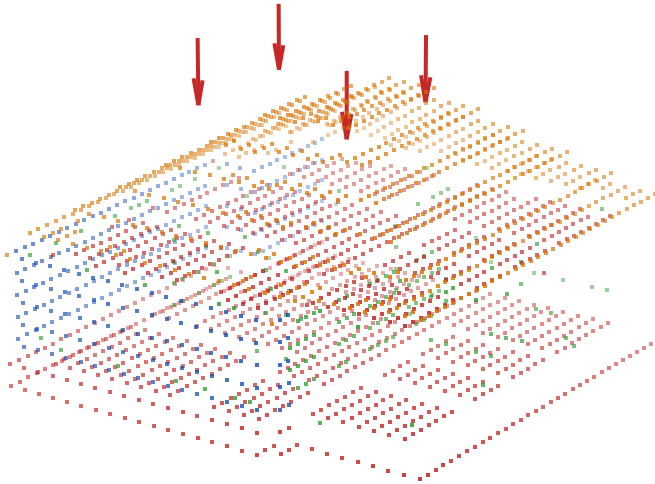
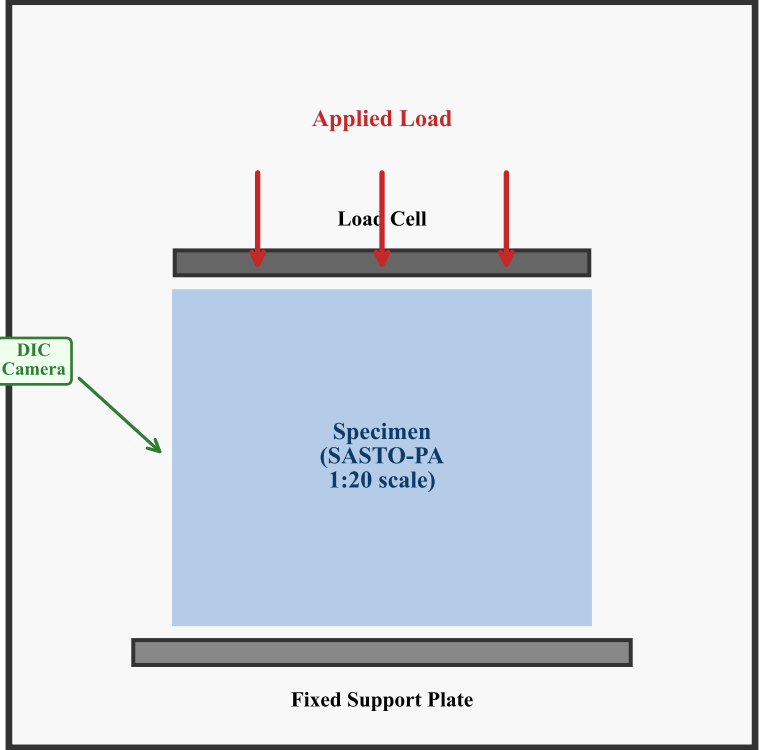


Physical Validation — 3D-Print Test Protocol (Future Work)

(a) Test Specimen
(1:20 scale, SASTO-PA)



(b) Compression Test Setup
(ASTM C39 adapted)



(c) Validation Criteria

Parameter	Requirement
Failure load	$\sigma_{pred} \pm 20\%$
Crack pattern	Match FEA stress map
Max deformation	$u_{pred} \pm 15\%$
Load rate	0.25 MPa/s
DIC resolution	0.01 mm
Scale factor	1:20
Material	Structural concrete